5. Refrigeration and air-conditioning :

5.1 Vapour compression refrigeration cycle-cycle on p-H & T-s diagrams; eco-friendly refrigerants - R134a,123; Systems like evaporators, condensers, compressor, expansion devices. Simple vapour absorption systems.

5.2 Psychrometry-properties; processes; charts; sensible heating and cooling; humidification and dehumidification effective temperature; air-conditioning load calculation; simple duct design.

MEDICAL SCIENCE

PAPER–I

1. Human Anatomy—Applied anatomy including blood and nerve supply of upper and lower limbs and joints of shoulder, hip and knee.

Gross anatomy, blood supply and lymphatic drainage of tongue, thyroid, mammary gland, stomach, liver, prostate, gonads and uterus

Applied anatomy of diaphragm, perineum and inguinal region.

Clinical anatomy of kidney, urinary bladder, uterine tubes, vas deferens.

Embryology—Placenta and placental barrier. Development of heart, gut, kidney, uterus, ovary, testis and their common congenital abnormalities.

Central and peripheral autonomic nervous system—Gross and clinical anatomy of ventricles of brain, circulation of cerebrospinal fluid; Neural pathways and lesions of cutaneous sensations, hearing and vision; Cranial nerves, distribution and clinical significance; Components of autonomic nervous system.

2. Human Physiology—Conduction and transmission of impulse, mechanism of contraction, neuromuscular transmission, reflexes, control of equilibrium, posture and muscle tone, descending pathways, functions of cerebellum, basal ganglia, Physiology of sleep and consciousness.

Endocrine system—Mechanism of action of hormones, formation, secretion, transport, metabolism, function and regulation of secretion of pancreas and pituitary gland.

Physiology of reproductive system—Menstrual cycle, lactation, pregnancy.

Blood—Development, regulation and fate of blood cells.

Cardio-vascular, cardiac output, blood pressure, regulation of cardiovascular functions;

3. Biochemistry—Organ function tests-liver, kidney, thyroid Protein synthesis.

Vitamins and minerals.

Restriction fragment length polymorphism (RFLP).

Polymerase chain reaction (PCR).

Radio-immunoassays (RIA).

4. Pathology—Inflammation and repair, disturbances of growth and cancer, Pathogenesis and histopathology of rheumatic and ischemic heart disease and diabetes mellitus. Differentiation between benign, malignant, primary and metastatic malignancies, Pathogenesis and histopathology of bronchogenic carcinoma, carcinoma breast, oral cancer, cancer cervix, leukemia, Etiology, pathogenesis and histopathology of-cirrhosis liver, glomerulonephritis, tuberculosis, acute osteomyelitis.

5. Microbiology—Humoral and cell mediated immunity. Diseases caused by and laboratory diagnosis of :

* Meningococcus, Salmonella
* Shigella, Herpes, Dengue, Polio
* HIV/AIDS, Malaria, E. Histolytica, Giardia
* Candida, Cryptococcus, Aspergillus

6. Pharmacology—Mechanism of action and side effects of the following drugs :

* Antipyretics and analgesics, Antibiotics, Antimalaria; Antikala-azar,Antidiabetics
* Antihypertensive, Antidiuretics, General and cardiac vasodilators, Antiviral, Antiparasitic, Antifungal, Immunosuppressants
* Anticancer

7. Forensic Medicine and Toxicology—Forensic examination of injuries and wounds; Examination of blood and seminal stains; poisoning, sedative overdose, hanging, drowning, burns, DNA and finger print study.

PAPER–II

1. General Medicine—Etiology, clinical features, diagnosis and principles of management (including prevention) of—Tetanus, Rabies, AIDS, Dengue, Kala-azar, Japanese Encephalitis.

Etiology, clinical features, diagnosis and principles of management of—Ischaemic heart disease, pulmonary embolism.

Bronchial asthma.

Pleural effusion, tuberculosis, Malabsorption syndromes, acid peptic diseases, Viral hepatitis and cirrhosis of liver.

Glomerulonerphritis and pyelonephritis, renal failure, nephrotic syndrome, renovascular hypertension, complications of diabetes mellitus, coagulation disorders, leukemia, Hypo and hyper thyrodism, meningitis and encephalitis.